Algebra & Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solving linear equations**

Solve $3x - 7 = 11$.

$3x = 18$, so $x = 6$.

2 **[Foundational]** • **Target: Solving linear equations**

Solve $5(x - 2) = 3x + 4$.

$5x - 10 = 3x + 4 \Rightarrow 2x = 14 \Rightarrow x = 7$.

3 **[Foundational]** • **Target: Solving linear equations**

Solve $\frac{3x - 2}{4} = x - 1$.

$3x - 2 = 4(x - 1) = 4x - 4 \Rightarrow -x = -2 \Rightarrow x = 2$.

4 **[Foundational]** • **Target: Solving linear equations**

Solve $2(x + 3) - 5 = 3(x - 1)$.

$2x + 1 = 3x - 3 \Rightarrow -x = -4 \Rightarrow x = 4$.

5 **[Foundational]** • **Target: Expanding and simplifying expressions**

Expand and simplify $3(2x - 1) - 2(x - 4)$.

$6x - 3 - 2x + 8 = 4x + 5$.

6 **[Foundational]** • **Target: Expanding and simplifying expressions**

Expand $(x + 5)(x - 2)$.

$x^2 - 2x + 5x - 10 = x^2 + 3x - 10$.

7 **[Foundational]** • **Target: Expanding and simplifying expressions**

Expand $(2x - 3)^2$.

$(2x - 3)^2 = 4x^2 - 12x + 9$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Expanding and simplifying expressions

Simplify $\frac{6x^2 - 9x}{3x}$ (for $x \neq 0$).

$\frac{3x(2x-3)}{3x} = 2x - 3$.

9 [Intermediate] • Target: Factoring quadratic expressions

Factor $x^2 + 7x + 12$.

$(x + 3)(x + 4)$.

10 [Intermediate] • Target: Factoring quadratic expressions

Factor $x^2 - 2x - 15$.

$(x - 5)(x + 3)$.

11 [Intermediate] • Target: Factoring quadratic expressions

Factor $2x^2 + 5x - 3$.

$(2x - 1)(x + 3)$.

12 [Intermediate] • Target: Factoring quadratic expressions

Factor $x^2 - 49$ using the difference of squares.

$(x - 7)(x + 7)$.

13 [Intermediate] • Target: Solving quadratic equations by factoring

Solve $x^2 - 5x + 6 = 0$.

$(x - 2)(x - 3) = 0$, so $x = 2$ or $x = 3$.

14 [Intermediate] • Target: Solving quadratic equations by factoring

Solve $x^2 + 2x - 8 = 0$.

$(x + 4)(x - 2) = 0$, so $x = -4$ or $x = 2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 8.01 Exponentials & Logarithms

Solve $2x^2 - 5x - 3 = 0$.

$(2x + 1)(x - 3) = 0$, so $x = -\frac{1}{2}$ or $x = 3$.

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16 [Challenging] • Target: The quadratic formula and the discriminant • Cross-Lesson: 9.01 Statistics & Probability

Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.

$a = 1, b = 3, c = 1$. $x = \frac{-3 \pm \sqrt{9 - 4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$.

17 [Challenging] • Target: The quadratic formula and the discriminant • Cross-Lesson: 2.01 Functions & Transformations

Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.

$\Delta = (-4)^2 - 4(2)(5) = 16 - 40 = -24$. Since $\Delta < 0$, there are no real roots.

18 [Challenging] • Target: The quadratic formula and the discriminant • Cross-Lesson: 3.01 Sequences & Series

Use the quadratic formula to solve $3x^2 - 2x - 1 = 0$.

$\Delta = (-2)^2 - 4(3)(-1) = 4 + 12 = 16$. $x = \frac{2 \pm 4}{6}$, so $x = 1$ or $x = -\frac{1}{3}$.

19 [Challenging] • Target: Absolute value equations • Cross-Lesson: 4.01 Trigonometry

Solve $|2x + 3| = 7$.

$2x + 3 = 7 \Rightarrow x = 2$, or $2x + 3 = -7 \Rightarrow x = -5$.

20 [Challenging] • Target: Linear inequalities • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Solve $2x - 5 < 9$ and graph the solution set on a number line.

$2x < 14 \Rightarrow x < 7$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Absolute value equations

The graph shows $y = |x - 3|$. At what x-value does the graph reach its minimum?

a. $x = 1$

b. $x = 3$ ✓

c. $x = -3$

d. $x = 0$

The vertex of the V-shape (the minimum) is at $x = 3$.

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22 [Foundational] • Target: Linear inequalities

Solve $3x - 7 < 14$, and identify the graph of its solution set.

a. $x < 7$ ✓

b. $x \geq 7$

c. $x \leq 7$

d. $x > 7$

$3x < 21 \Rightarrow x < 7$.

23 [Foundational] • Target: Linear inequalities

Solve $-4x + 6 \geq -10$, and identify the graph of its solution set.

a. $x > 4$

b. $x < 4$

c. $x \leq 4$ ✓

d. $x \geq 4$

$-4x \geq -16$; dividing by -4 flips the sign: $x \leq 4$.

24 [Foundational] • Target: Linear-quadratic systems

The graph shows the system $y = x^2 - 4$, $y = 3x$. What are the points of intersection?

a. $(4, 12)$ and $(1, 3)$

b. $(-4, -12)$ and $(1, 3)$

c. $(-1, -3)$ and $(1, 3)$

d. $(4, 12)$ **and** $(-1, -3)$ ✓

The curve and line cross at $(-1, -3)$ and $(4, 12)$.

25 [Foundational] • Target: Linear-quadratic systems

The graph shows the system $y = x^2$, $y = 2x + 3$. What are the points of intersection?

a. $(3, 9)$ and $(1, 1)$

b. $(-1, 1)$ and $(1, 1)$

c. $(-3, 9)$ and $(1, 1)$

d. $(3, 9)$ **and** $(-1, 1)$ ✓

The curve and line cross at $(-1, 1)$ and $(3, 9)$.

26 [Foundational] • Target: Solving linear equations

Solve $5x + 3 = 2x - 9$.

a. $x = -2$

b. $x = 2$

c. $x = 4$

d. $x = -4$ ✓

$3x = -12 \Rightarrow x = -4$.

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27 [Foundational] • Target: Factoring quadratic expressions

Factor $2x^2 - x - 6$.

a. $(2x + 3)(x - 2)$ ✓

b. $(2x - 3)(x + 2)$

c. $(x + 3)(2x - 2)$

d. $(2x + 1)(x - 6)$

$(2x + 3)(x - 2) = 2x^2 - 4x + 3x - 6 = 2x^2 - x - 6$.

28 [Foundational] • Target: The quadratic formula and the discriminant

Use the quadratic formula to solve $x^2 + 5x + 2 = 0$, leaving the answer in surd form.

a. $x = \frac{5 \pm \sqrt{17}}{2}$

b. $x = \frac{-5 \pm \sqrt{17}}{4}$

c. $x = \frac{-5 \pm \sqrt{33}}{2}$

d. $x = \frac{-5 \pm \sqrt{17}}{2}$ ✓

Discriminant $= 25 - 8 = 17$; $x = \frac{-5 \pm \sqrt{17}}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The quadratic formula and the discriminant

Use the quadratic formula to solve $2x^2 + 3x - 2 = 0$.

a. $x = 1$ or $x = -2$

b. $x = -0.5$ or $x = 2$

c. $x = 0.5$ **or** $x = -2$ ✓

d. $x = 0.5$ or $x = 2$

Discriminant $= 9 + 16 = 25$; $x = \frac{-3 \pm 5}{4}$: $x = 0.5$ or $x = -2$.

30 [Intermediate] • Target: Linear inequalities

Solve $2x + 5 > -3$ and $x - 1 < 6$ together (both must hold). Find the combined solution set.

a. $x > -4$

b. $-4 < x \leq 7$

c. $x < 7$

d. $-4 < x < 7$ ✓

$2x + 5 > -3 \Rightarrow x > -4$; $x - 1 < 6 \Rightarrow x < 7$. Together: $-4 < x < 7$.

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31 [Intermediate] • Target: Systems of linear equations

The sum of two numbers is 22 and their difference is 6. Find the two numbers.

a. 11 and 11

b. 14 and 6

c. 16 and 6

d. 14 **and** 8 ✓

$x + y = 22, x - y = 6 \Rightarrow x = 14, y = 8.$

32 [Intermediate] • Target: Systems of linear equations

Solve the system $2x - y = 7$, $3x + 2y = 21$.

a. $x = 5, y = -3$

b. $x = 5$, $y = 3$ ✓

c. $x = 3, y = 5$

d. $x = 7, y = 7$

$y = 2x - 7 \Rightarrow 3x + 2(2x - 7) = 21 \Rightarrow 7x = 35 \Rightarrow x = 5$; $y = 3$.

33 [Intermediate] • Target: Rational equations

Solve $\frac{4}{x-1} = \frac{2}{x+2}$.

a. $x = -5$ ✓

b. $x = 5$

c. $x = -1$

d. $x = 2$

Cross-multiply: $4(x + 2) = 2(x - 1) \Rightarrow 4x + 8 = 2x - 2 \Rightarrow x = -5$.

34 [Intermediate] • Target: Rational equations

Solve $\frac{6}{x} = \frac{2}{x-4}$.

a. $x = 12$

b. $x = 4$

c. $x = -6$

d. $x = 6$ ✓

Cross-multiply: $6(x - 4) = 2x \Rightarrow 6x - 24 = 2x \Rightarrow x = 6$.

35 [Intermediate] • Target: Solving for a variable in a formula

The perimeter of a rectangle is $P = 2l + 2w$. Solve for l.

a. $l = \frac{P}{2} - w$

b. $l = \frac{P-2w}{2}$ ✓

c. $l = \frac{P+2w}{2}$

d. $l = P - 2w$

$P - 2w = 2l \Rightarrow l = \frac{P-2w}{2}$ (equivalently $\frac{P}{2} - w$).

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36 [Intermediate] • Target: Solving for a variable in a formula

The formula for simple interest is $I = Prt$. Solve for r.

a. $r = \frac{I}{P}$

b. $r = \frac{I}{Pt}$ ✓

c. $r = \frac{Pt}{I}$

d. $r = I - Pt$

Dividing both sides by Pt: $r = \frac{I}{Pt}$.

37 [Intermediate] • Target: Solving for a variable in a formula

The formula $F = \frac{9}{5}C + 32$ converts Celsius to Fahrenheit. Solve for C.

a. $C = \frac{5(F-32)}{9}$ ✓

b. $C = \frac{5F-32}{9}$

c. $C = \frac{5}{9}F + 32$

d. $C = \frac{9(F-32)}{5}$

$F - 32 = \frac{9}{5}C \Rightarrow C = \frac{5(F-32)}{9}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving for a variable in a formula • Cross-Lesson: 3.01 Sequences & Series

The volume of a cylinder is $V = \pi r^2 h$. Solve for h.

a. $h = \frac{V}{\pi r^2}$ ✓

b. $h = V - \pi r^2$

c. $h = \frac{V}{\pi r}$

d. $h = \frac{\pi r^2}{V}$

Dividing both sides by πr^2: $h = \frac{V}{\pi r^2}$.

39 [Challenging] • Target: Linear-quadratic systems • Cross-Lesson: 4.01 Trigonometry

Solve the system $y = x^2 - 4$, $y = 3x$.

a. $(-4, -12)$ and $(1, 3)$

b. $(-1, -3)$ and $(1, 3)$

c. $(4, 12)$ and $(1, 3)$

d. $(4, 12)$ **and** $(-1, -3)$ ✓

$x^2 - 4 = 3x \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x-4)(x+1) = 0$: $x = 4, y = 12$ or $x = -1, y = -3$.

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40 [Challenging] • Target: Linear-quadratic systems • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Solve the system $y = x^2$, $y = 2x + 3$.

a. $(-3, 9)$ and $(1, 1)$

b. $(3, 9)$ and $(1, 1)$

c. $(-1, 1)$ and $(1, 1)$

d. $(3, 9)$ **and** $(-1, 1)$ ✓

$x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$: $x = 3, y = 9$ or $x = -1, y = 1$.

41 [Challenging] • Target: Word problems: setting up and solving equations • Cross-Lesson: 6.01 Limits & Derivatives

A rectangular garden has a length that is 5 meters more than its width, and its area is 84 square meters. Find the width.

a. 12

b. 7 ✓

c. 9

d. 14

$w(w + 5) = 84 \Rightarrow (w + 12)(w - 7) = 0$; the positive root is $w = 7$.

42 [Challenging] • Target: Word problems: setting up and solving equations • Cross-Lesson: 7.01 Integration & Applications

A mobile-data plan charges a fixed monthly fee of 40 riyals plus 2 riyals per GB used. If a user's bill was 68 riyals, how many GB did they use?

a. 34

b. 14 ✓

c. 54

d. 28

$40 + 2g = 68 \Rightarrow 2g = 28 \Rightarrow g = 14$.

43 [Challenging] • Target: Algebra & Equations • Cross-Lesson: 8.01 Exponentials & Logarithms

Solve $4(x - 1) = x + 11$.

a. -5

b. 3

c. 5 ✓

d. 7

$4x - 4 = x + 11 \Rightarrow 3x = 15 \Rightarrow x = 5$.

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44 [Challenging] • Target: Algebra & Equations • Cross-Lesson: 9.01 Statistics & Probability

Solve $4(x + 9) = x + 36$.

a. 3

b. -2

c. 0 ✓

d. 2

$4x + 36 = x + 36 \Rightarrow 3x = 0 \Rightarrow x = 0$.

45 [Challenging] • Target: Algebra & Equations • Cross-Lesson: 2.01 Functions & Transformations

Solve $4(x - 1) = 2x - 8$.

a. 2

b. 0

c. -4

d. -2 ✓

$4x - 4 = 2x - 8 \Rightarrow 2x = -4 \Rightarrow x = -2$.

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